

```
import os

print(os.listdir('/content'))

['.config', 'train.csv', 'sample_data']
```

```
['sample_data', 'train.csv']
```

```
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```
['sample_data', 'train (1).csv']
```

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```

```
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```

```
import pandas as pd

# Read dataset with encoding fix
data = pd.read_csv('/content/train.csv', encoding='latin1')

# Show first 5 rows
data.head()
```

	textID	text	selected_text	sentiment	Time of Tweet	Age of User	Country
0	cb774db0d1	I'd have responded, if I were going	I'd have responded, if I were going	neutral	morning	0-20	Afghanistan
1	549e992a42	Sooo SAD I will miss you here in San Diego!!!	Sooo SAD	negative	noon	21-30	Albania
2	088c60f138	my boss is bullying me...	bullying me	negative	night	31-45	Algeria
3	9642c003ef	what interview! leave me alone	leave me alone	negative	morning	46-60	Andorra
4	358bd9e861	Sons of ****, why couldn't they put them on t...	Sons of ****,	negative	noon	60-70	Angola

Next steps:

[Generate code with data](#)

[New interactive sheet](#)

Dataset information
data.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 27481 entries, 0 to 27480
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   textID                 27481 non-null  object
1   text                   27480 non-null  object
2   selected_text          27480 non-null  object
3   sentiment               27481 non-null  object
4   Time of Tweet          27481 non-null  object
5   Age of User            27481 non-null  object
6   Country                27481 non-null  object
7   Population -2020       27481 non-null  int64
8   Land Area (Km²)        27481 non-null  float64
9   Density (P/Km²)        27481 non-null  int64
dtypes: float64(1), int64(2), object(7)
memory usage: 2.1+ MB
```

```
# Missing values
data.isnull().sum()
```

	0
textID	0
text	1
selected_text	1
sentiment	0
Time of Tweet	0
Age of User	0
Country	0
Population -2020	0
Land Area (Km ²)	0
Density (P/Km ²)	0

```
dtype: int64
```

```
# Column names
data.columns
```

```
Index(['textID', 'text', 'selected_text', 'sentiment', 'Time of Tweet',
      'Age of User', 'Country', 'Population -2020', 'Land Area (Km²)',
      'Density (P/Km²)'],
      dtype='object')
```

```
# Statistical summary
data.describe()
```

	Population -2020	Land Area (Km ²)	Density (P/Km ²)
count	2.748100e+04	2.748100e+04	27481.000000
mean	4.018497e+07	6.621730e+05	357.686583
std	1.504946e+08	1.807425e+06	2013.750702
min	8.010000e+02	0.000000e+00	2.000000
25%	1.968001e+06	2.281000e+04	35.000000
50%	8.655535e+06	1.118900e+05	89.000000
75%	2.843594e+07	5.279700e+05	214.000000
max	1.439324e+09	1.637687e+07	26337.000000



```
# Remove missing values
data = data.dropna()
```

```
import re

def clean_text(text):
    text = str(text)
    text = text.lower()
    text = re.sub(r'^a-zA-Z ', '', text)
    return text
```

```
data.columns
```

```
Index(['textID', 'text', 'selected_text', 'sentiment', 'Time of Tweet',
      'Age of User', 'Country', 'Population -2020', 'Land Area (Km²)',
      'Density (P/Km²)'],
      dtype='object')
```

```
data['clean_text'] = data['text'].apply(clean_text)

data.head()
```

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```
from collections import Counter

all_words = ' '.join(data['clean_text']).split()

word_count = Counter(all_words)

print(word_count.most_common(10))
```

```
[('i', 13179), ('to', 10015), ('the', 8984), ('a', 6714), ('my', 5512), (
```

```
import matplotlib.pyplot as plt

words = dict(word_count.most_common(10))

plt.bar(words.keys(), words.values())
plt.xticks(rotation=45)
plt.show()
```

